



Elham Rivaz

// Computer Engineering Graduate

AI/ML Engineer | AI Research Intern @ University of Queensland Bachelor

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> PROFESSIONAL SUMMARY

Computer Engineering graduate specializing in AI/ML, with hands-on experience in deep learning and computer vision through applied research and industry-focused projects. Proven ability to boost model performance significantly, demonstrated by improving classification accuracy from 58.57% to 97.86% in a binary image classification project. Combines strong analytical thinking with fast execution and effective teamwork to turn AI concepts into practical, high-impact solutions.

> SKILLS

AI/ML: Deep Learning | CNNs | Transfer Learning | Ensemble Methods | Feature Fusion

LLM/GenAI: RAG Systems | LangChain | ChromaDB | OpenRouter | Prompt Engineering

Languages: Python | C++ | C | C# | SQL

Frameworks: Django | PyTorch

Data: Pandas | Jupyter | MySQL | Data Structures

Core CS: Algorithms | Operating Systems | Networking | Computer Architecture

Mathematics: Linear Algebra | Numerical Analysis | Probability & Statistics | Discrete Math

Soft Skills: Teamwork | Problem Solving

> EXPERIENCE

AI Research Intern (Remote) @ University of Queensland

2024-09 – 2025-01

Worked on deep learning and computer vision projects focused on Convolutional Neural Networks (CNNs). Developed a binary image classification system using four different model approaches (simple CNN, Ensemble Stacking, Transfer Learning, and Feature Fusion), improving accuracy from 58.57% to 97.86%.

Tech Stack: Python | Deep Learning | CNN | Transfer Learning

- § - Improved model accuracy from 58.57% to 97.86% through iterative experimentation across four architectures
- § - Implemented and compared 4 approaches: simple CNN, Ensemble Stacking, Transfer Learning, and Feature Fusion
- § - Applied deep learning techniques to real-world image classification problems

Teaching Assistant (TA) @ Shiraz University

2023-11 – 2024-02

Served as teaching assistant for Fundamentals of Computer Science using Python. Co-taught course material, resolved student questions, administered exams, and handled grading.

Technologies: Python

[Tech Stack: Python](#) | [Teaching](#) | [Debugging](#) | [Exam Design](#)

- \$ - Assisted in teaching core computer science concepts to undergraduate students
- \$ - Managed exam administration and grading for the course
- \$ - Provided one-on-one support to help students debug and understand code

Central Council Member @ IT Scientific Association, Shiraz University

2023-10 – 2024-10

Active member of the central council of the IT Student Scientific Association, contributing to the organization of multiple courses and events, including two major conferences on Artificial Intelligence and DevOps, as well as addressing student affairs and concerns.

[Tech Stack: Event Management](#) | [Team Leadership](#) | [Public Speaking](#) | [Community Engagement](#)

- \$ - Co-organized two major conferences on AI and DevOps
- \$ - Coordinated multiple training courses and student events
- \$ - Represented and addressed concerns of the student community

> PROJECTS

Binary Image Classifier (Cat vs. Dog)

2024-01 – 2024-12

Developed a binary image classification system (cat vs. dog) using four progressive model versions — simple CNN, Ensemble Stacking, Transfer Learning, and Feature Fusion — improving accuracy from 58.57% to 97.86%.

[Tech Stack: Python](#) | [PyTorch](#) | [CNN](#) | [Transfer Learning](#) | [Deep Learning](#)

- \$ - Achieved 97.86% accuracy through iterative architecture improvements
- \$ - Compared 4 distinct modeling approaches on the same task

RAG Document Q&A System

2025-01 – 2025-12

Built a full-stack Retrieval-Augmented Generation (RAG) system that allows users to upload documents and receive AI-generated answers with source citations. Supports multilingual search and exposes a complete REST API.

[Tech Stack: Django](#) | [LangChain](#) | [ChromaDB](#) | [OpenRouter](#) | [Python](#)

- \$ - Implemented multilingual document search and retrieval
- \$ - Delivered cited, source-grounded answers (reduces hallucination)
- \$ - Exposed a complete REST API for integration

CS50AI Projects — 12 Mini AI Projects

2024-01 – 2024-12

Completed 12 mini projects as part of Harvard's CS50AI course, covering Search, Knowledge, Uncertainty, Optimization, Machine Learning, Deep Learning, and NLP. Earned Silver Medal in CS50AI Puzzle Day.
Technologies: Python, Search Algorithms, Machine Learning, NLP, Deep Learning
[Tech Stack: Python](#) | [Search Algorithms](#) | [Machine Learning](#) | [NLP](#) | [Deep Learning](#)

- \$ - Completed hands-on projects across 7 core AI domains
- \$ - Awarded Silver Medal in CS50AI Puzzle Day (Harvard University)

The Lost Sequence (Adventure Game)

2024-01 – 2024-12

Solo-developed a mystery adventure game combining creativity with Generative AI capabilities, including AI-generated imagery, AI-generated voice/audio, and AI-assisted code development.
Technologies: Generative AI, Python, Game Design
[Tech Stack: Generative AI](#) | [Python](#) | [Game Design](#)

- \$ - Independently designed and built a full narrative adventure game
- \$ - Integrated multiple Generative AI tools (image, voice, code generation)

Dictionary for Indexer

2024-01 – 2024-12

Group project focused on designing and developing a dictionary module for the indexer component of an information retrieval system.
[Tech Stack: Python](#) | [Data Structures](#) | [Information Retrieval](#)
\$ - Collaborated in a team to build a core indexing component
\$ - Applied information retrieval concepts in a practical system design

> GITHUB PROJECTS

ByElham/binary-image-classifier

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[Python](#)

Binary image classification system (cat vs. dog) achieving 97.86% accuracy across 4 progressive model architectures — simple CNN, Ensemble Stacking, Transfer Learning, and Feature Fusion.
Technologies: Python, PyTorch, CNN, Deep Learning

ByElham/studybuild-data-cleaning-01

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[Jupyter Notebook](#)

Practiced end-to-end data cleaning workflow on a real-world dataset — handling missing values, duplicates, inconsistent formatting, and outliers.
Technologies: Python, Jupyter Notebook, Pandas

[Jupyter Notebook](#)

Analyzed customer behavior data as part of a community project, generating business insights and data-driven recommendations using Python.

Technologies: Python, Jupyter Notebook, Data Analysis

> EDUCATION

Bachelor's - Computer Engineering (Software) — Shiraz University

2021-09 – 2025-09

GPA: 19.20/20

§ - Ranked #1 among 72+ students in the Computer Engineering program (June 2025)

§ - Top Student Award in English and German for 6 consecutive semesters

> HONORS & AWARDS

Ranked #1 Among 72+ Students

Jun 2025

Top-ranked student in the Computer Engineering undergraduate program at Shiraz University among all peers in the same cohort.

EPFL Switzerland Internship Scholarship — Final Shortlist

Feb 2024

Selected for the final shortlist of a highly competitive internship scholarship at EPFL (École Polytechnique Fédérale de Lausanne), Switzerland.

CS50x Puzzle Day — Silver Medal

Apr 2024

Awarded Silver Medal in Harvard University's CS50 Puzzle Day competition.

Top Student Award — English & German

— (6 consecutive semesters)

Recognized as Top Student in English and German language courses for 6 consecutive semesters.

> CERTIFICATIONS

CS50AI — Harvard University

Oct 2025

Machine Learning Specialization — Stanford University (Andrew Ng)

Generative AI (5-Day Gen AI Intensive) — Google & Kaggle

Mar 2025

AI Agents: Intensive Vibe Coding Course With Google — Kaggle

Aug 2026

Applied Machine Learning (Project-based) — Maktabkhooneh

May 2026

Comprehensive Machine Learning Course — Sharif University of Technology

May 2026

AI Mathematics (Linear Algebra, Statistics & Probability) — Shiraz University

May 2024

Soft Skills Development — Menteeno

Sep 2025

ICDL — Shiraz University

Feb 2022

Fundamentals of C# — SIMI (South Region Industrial Management Organization)

Sep 2019

> RESEARCH

Medical Image Fusion

May 2025

Contributed to a research project aimed at improving medical image fusion techniques for enhanced clinical analysis, focusing on proposing methodology improvements and dataset collection.
Supervisor: Moloud Abdar.

> LANGUAGES

Persian: Native

English: Intermediate

German: Intermediate

Arabic: Basic

> QR CODE



CS50AI Certificate